

Omni Directional Wheel base Control

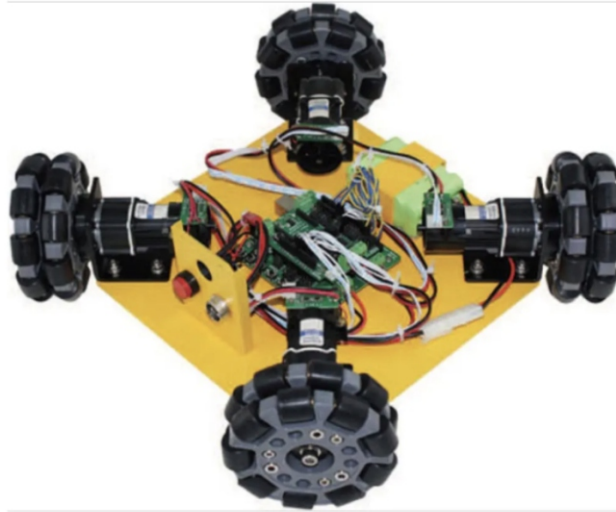


Figure 01 - Omnidirectional wheel base

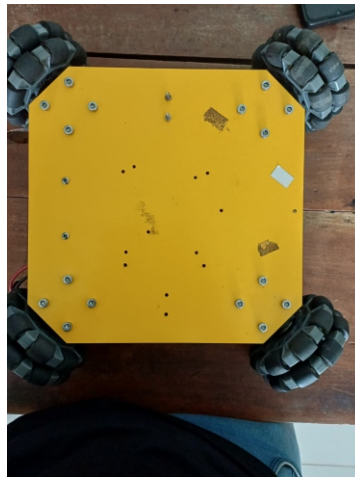


Figure 02/ 03/ 04 - Checking of the omnidirectional wheel base

Code used for Initial Control Checking:

```
#INCLUDE <PINCHANGEINT.H>
#include <PINCHANGEINTCONFIG.H>
#include <EEPROM.H>
#include <FUZZY_TABLE.H>
#include <PID_BETA6.H>
#include <MOTORWHEEL.H>
#include <OMNI4WD.H>

#include <FUZZY_TABLE.H>
#include <PID_BETA6.H>
```

```
/*
```

```
    \           /  
WHEEL1 \       / WHEEL4  
LEFT   \      /  RIGHT
```

```
POWER SWITCH
```

```
    /           \  
WHEEL2 /       \ WHEEL3  
RIGHT  /      \  LEFT
```

```
*/
```

```
/*
```

```
IRQISR(IRQ1,ISR1);  
MOTORWHEEL WHEEL1(5,4,12,13,&IRQ1);
```

```
IRQISR(IRQ2,ISR2);  
MOTORWHEEL WHEEL2(6,7,14,15,&IRQ2);
```

```
IRQISR(IRQ3,ISR3);  
MOTORWHEEL WHEEL3(9,8,16,17,&IRQ3);
```

```
IRQISR(IRQ4,ISR4);  
MOTORWHEEL WHEEL4(10,11,18,19,&IRQ4);  
*/
```

```
IRQISR(IRQ1,ISR1);  
MOTORWHEEL WHEEL1(3,2,4,5,&IRQ1);
```

```
IRQISR(IRQ2,ISR2);  
MOTORWHEEL WHEEL2(11,12,14,15,&IRQ2);
```

```
IRQISR(IRQ3,ISR3);  
MOTORWHEEL WHEEL3(9,8,16,17,&IRQ3);
```

```
IRQISR(IRQ4,ISR4);  
MOTORWHEEL WHEEL4(10,7,18,19,&IRQ4);
```

```
OMNI4WD OMNI(&WHEEL1,&WHEEL2,&WHEEL3,&WHEEL4);
```

```
VOID SETUP() {  
//TCCROB=TCCROB&0xF8|0x01; // WARNING!! IT WILL CHANGE MILLIS()  
TCCR1B=TCCR1B&0xF8|0x01; // PIN9,PIN10 PWM 31250Hz  
TCCR2B=TCCR2B&0xF8|0x01; // PIN3,PIN11 PWM 31250Hz
```

```
OMNI.PIDENABLE(0.31,0.01,0,10);
```

```
}
```

```
VOID LOOP() {  
OMNI.DEMOActions(200,5000,500,FALSE);
```

```
}
```